2025

AUSTRALIAN MATHEMATICS COMPETITION

AMC.

AUSTRALIAN MATHS TRUST

Level A
Middle Primary
Grades 3 – 5

DATE
27 Sept.

TIME ALLOWED
60 minutes

Information 考试信息

- There are 30 questions: 25 multiple-choice and 5 integer.
  本次考试共 30 题，包含 25 道选择题和 5 道答案为整数的填空题。
- You are allowed 60 minutes to do the questions. There is no extra reading time.
  考试时长为 60 分钟，无额外读题时间。
- There are no penalties for incorrect answers. You should attempt all questions.
  答错不扣分，鼓励学生尝试解答所有问题。
- The questions have been checked thoroughly. Each question is clearly written and no further explanation will be given.
  本场考试的所有问题已仔细核查。题目陈述清晰，不提供更多解释。
- Diagrams are not drawn to scale. They are intended only as aids.
  图形未按比例绘制，仅供参考。

Competition rules 考试规则

- You must maintain silence at all times.
  考试过程中请保持安静。
- Do not open the paper until told to do so.
  收到监考老师指示前，请勿打开试卷。
- Mobile phones and smart watches are not allowed. If you have one you must give it to a supervisor, who will store it in a safe place.
  禁止使用手机和智能手表。如有携带这些设备，请立刻交与监考老师保管。
- You may use printed language dictionaries.
  允许使用纸质语言词典。
- You may use calculators.
  允许使用计算器。
- You may not borrow equipment without a supervisor's permission.
  未经监考老师许可，禁止向他人借用工具。
- You must not leave your seat during the competition. You must remain seated until the competition finishes.
  考试期间禁止离开座位，直到考试结束。
- If you wish to leave the room, a supervisor must accompany you. Raise your hand and wait for a supervisor to come to you.
  如需离开考场，必须举手示意并等待监考老师陪同。
- If you have any other questions or problems, raise your hand and wait for a supervisor.
  如果有其他问题，请举手示意监考老师。

No student may sit the competition more than once or sit more than one division of the competition. AMT conducts integrity checks on competition results and reserves the right to withhold results or disqualify students if plagiarism or duplicate sittings are suspected.

IMPORTANT

All papers must be immediately returned to a supervisor at the end of the competition.
考试结束后，由监考老师回收所有试卷。

This paper is not for circulation or distribution until Wednesday 1 October 2025.
在 2025 年 10 月 1 日星期三之前禁止讨论或传播试卷内容。

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Middle Primary Division

Questions 1 to 10, 3 marks each
1-10 题，每题 3 分

1. In the number sentence $16 - \triangle = 9$, what is the value of $\triangle$?
在算式 $16 - \triangle = 9$ 中，请问 $\triangle$ 代表的数字是多少？

(A) 3 (B) 5 (C) 7 (D) 9 (E) 16

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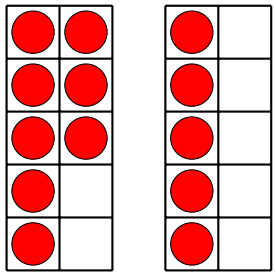
2. How many red counters have been placed on these tens-frames?
请问右侧的十格阵中放置了多少个红色计数圆片？

(A) 12 (B) 13 (C) 14 (D) 15 (E) 16



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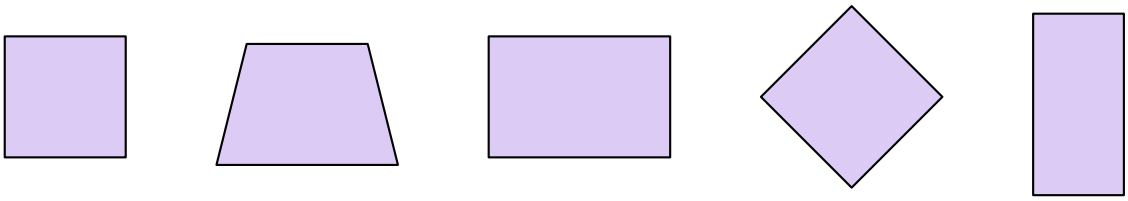
3. Which one of the following is **NOT** a rectangle?
请问以下哪个图形**不是**长方形？



(A) (B) (C) (D) (E)

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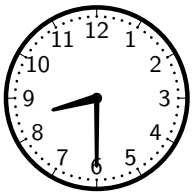
4. I am supposed to be at school at 8:45 am but this clock shows the time when I arrived. This means that I arrived
我计划在早上 8:45 到达学校，但当我到达学校时，
钟表显示的时间如右图所示。这说明我比计划的时间：



(A) 15 minutes early 早到 15 分钟
(B) 15 minutes late 晚到 15 分钟
(C) 45 minutes early 早到 45 分钟
(D) 45 minutes late 晚到 45 分钟
(E) Half an hour early 早到半小时

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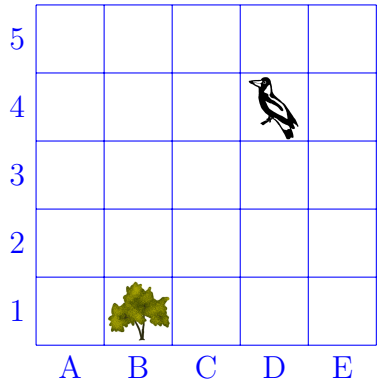
5. The bird in square D4 needs to fly back to the tree in square B1. Which one of the following could be the path taken?
位于 D4 格的小鸟需要飞回 B1 格的大树处。请问以下哪一条可能是它的飞行路线？



(A) 2 squares right and 2 squares down
向右飞 2 格，再向下飞 2 格

(B) 3 squares left and 1 square up
向左飞 3 格，再向上飞 1 格

(C) 3 squares right and 2 squares up
向右飞 3 格，再向上飞 2 格

(D) 1 square right and 3 squares down
向右飞 1 格，再向下飞 3 格

(E) 2 squares left and 3 squares down
向左飞 2 格，再向下飞 3 格

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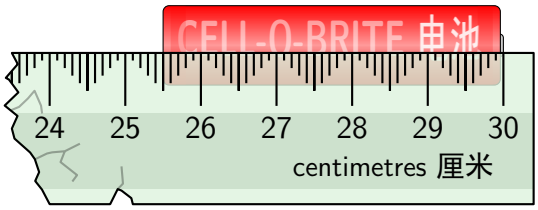
6. Pip's dog chewed her ruler, but she could use what was left to measure this battery. How long is the battery?
皮皮的小狗咬坏了她的直尺，但她可以用直尺剩下的部分测量一节电池的长度。请问这节电池的长度是多少？



(A) 40 mm (B) 45 mm (C) 50 mm (D) 55 mm (E) 255 mm

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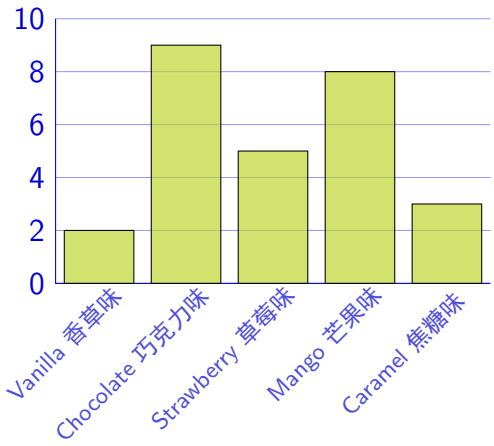
7. A teacher surveyed his students about their favourite ice cream flavours. The survey results are shown in this graph. How many students are in his class?
一位老师统计了学生们最喜欢的冰淇淋口味。统计结果如图所示。请问这个班级共有多少名学生？



(A) 5 (B) 9 (C) 15 (D) 21 (E) 27

8. I buy a \$4 bag of figs from a roadside stall. I have a \$5 note, a \$2 coin and a \$1 coin. In the money box, there is a \$10 note and two \$2 coins. Which one of the following pays \$4 for the figs?

我在路边摊买了一包标价为 4 美元的无花果。我身上有一张 5 美元纸币、一枚 2 美元硬币和一枚 1 美元硬币。收银台的钱箱里有一张 10 美元纸币和两枚 2 美元硬币。请问以下哪种方式可以支付 4 美元的无花果费用？

(A) Take the \$10 note out and put my \$5 note and \$1 coin in.
将 10 美元纸币从钱箱取出，并放入我的 5 美元纸币和 1 美元硬币。

(B) Put both of my coins in the box.
将我的两枚硬币全部放入钱箱。

(C) Take one \$2 coin out and put my \$5 note in.
将一枚 2 美元硬币从钱箱取出，并放入我的 5 美元纸币。

(D) Put all of my money in the box.
将我身上所有的钱都放入钱箱。

(E) Take a \$2 coin out and put my \$5 note and \$1 coin in.
将一枚 2 美元硬币从钱箱取出，并放入我的 5 美元纸币和 1 美元硬币。

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9. This year is 2025, which contains only 3 different digits: 0, 2 and 5. In how many years' time will the year next have **all 4** digits different?

今年是 2025 年，这个年份里只包含 3 个不同的数字：0、2 和 5。请问接下来再过多少年，才会出现 **4** 个数字**都**不相同的年份？

(A) 1 year　1 年　(B) 5 years　5 年　(C) 6 years　6 年　(D) 9 years　9 年　(E) 10 years　10 年

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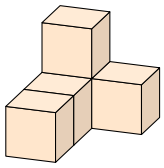
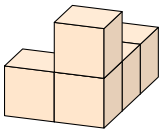
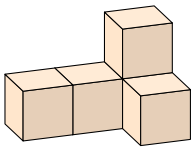
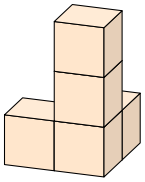
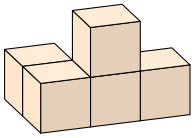
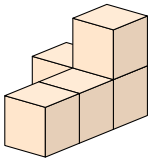
10. The solid shown here is made from 5 cubes.



Which one of the following five pictures is **not** a possible view of the solid?

右侧所示的立体图形由 5 个立方体组成。

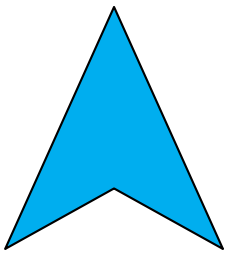
请判断在以下五个视图中，哪一个**不**可能是该立体图形的视图？

(A) 
(B) 
(C) 
(D) 
(E) 

Questions 11 to 20, 4 marks each
11-20 题，每题 4 分

11. This arrowhead shape has two equal short sides and two equal long sides. Its perimeter is 20 cm and each short side is 3 cm long.

此箭头图形由两条等长的短边和两条等长的长边组成。已知该图形的周长为 20 cm，且每条短边的长度为 3 cm。



How long is each long side?

请问每条长边的长度是多少？

(A) 4 cm (B) 5 cm (C) 6 cm (D) 7 cm (E) 8 cm

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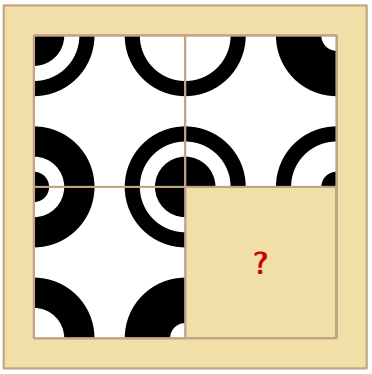
12. In this puzzle, the tiles have patterns on one side only. These tiles must be placed in the frame so that the patterns in the corners match.

在这个拼图游戏中，拼图块仅单面印有花纹。这些拼图块需要被放置在拼图相框中，使角落处的花纹能够相互匹配。



Which one of the following tiles can be used to finish the puzzle?

请问以下哪个选项可以用来完成这个拼图？

(A) (B) (C) (D) (E)

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13. Munda has 11 building blocks for his tower. Kaya has 3 times that many, and Bobby has 14 less than Kaya. How many blocks in total do Munda and Bobby have?

在搭塔游戏中，小蒙使用了 11 块积木，小凯使用的积木是小蒙的 3 倍，而小贝使用的积木比小凯少 14 块。请问小蒙和小贝一共使用了多少块积木？

(A) 26 (B) 28 (C) 30 (D) 32 (E) 34

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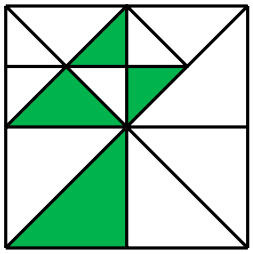
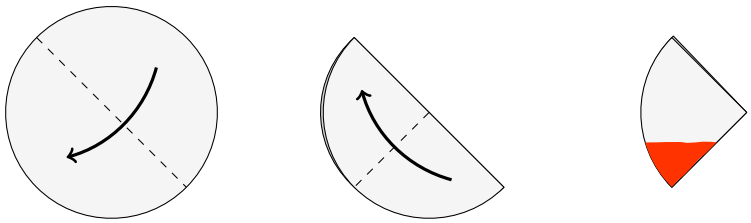
14. What fraction of this square is shaded?
请问阴影部分面积占整个正方形面积的几分之几？

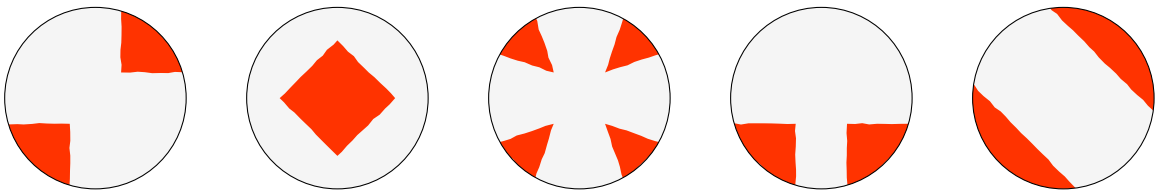


(A) One-half $\frac{1}{2}$ (B) One-third $\frac{1}{3}$ (C) One-quarter $\frac{1}{4}$
(D) One-fifth $\frac{1}{5}$ (E) One-eighth $\frac{1}{8}$

15. A circle of paper is folded in half and then in half again as shown. One corner is then dipped in red ink.
如图所示，将一张圆形纸片对折，然后再进行一次对折。接着，将折好的纸片的一个角浸入红色墨水中。



What does the circle look like when unfolded and laid flat?
请问将这张圆形纸片展开并平铺后，它看起来是什么样子的？



(A) (B) (C) (D) (E)

16. Timmy has a bag with 9 balls in it, numbered 1 to 9. He pulls one ball out at random and looks at the number. Which of the following is the most likely to occur?
小蒂的袋中有 9 颗球，编号从 1 至 9。他从中随机取出一颗球并查看上面的数字。请问以下哪种情况最有可能发生？

(A) The number is odd. 该数字为奇数
(B) The number is even. 该数字为偶数
(C) The number is less than 4. 该数字小于 4
(D) The number is more than 5. 该数字大于 5
(E) The number is a multiple of 3. 该数字是 3 的倍数

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17. It's raining and Joanna is one of 7 people trying to get from the house to the bus stop. The group has 3 umbrellas to keep them dry, but each is only big enough for one person.
下雨天，小乔和其他 6 个人（共 7 人）需要从家前往公交车站。他们一共有 3 把雨伞，但每把雨伞每次只能供 1 人使用。

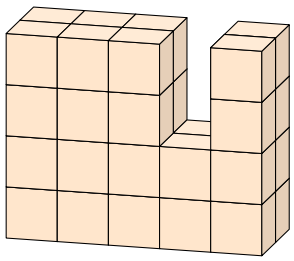
Joanna makes a plan to keep everyone dry where she walks between the house and the bus stop more than once, carrying the extra umbrellas if necessary, and everyone else just walks from the house to the bus stop with one umbrella.
为确保所有人不被淋湿，小乔制定了一个方案：她可以多次往返于家和公交车站之间运送雨伞，必要时可携带其余两把雨伞；而其他人则只需要每人持一把雨伞直接前往公交车站。

What is the smallest number of times that Joanna needs to walk from the house to the bus stop?
请问在该方案下，小乔从家到车站至少需要走多少次？

(A) 2 (B) 3 (C) 4 (D) 5 (E) 6

---

18. This is a view of a solid made by stacking 36 identical cubes. How many cubes are completely hidden in this view?
图中展示了一个由 36 个相同立方体堆叠而成的立体图形。请问在此视角下，有多少个立方体被完全遮挡？



(A) 6 (B) 7 (C) 8 (D) 9 (E) 10

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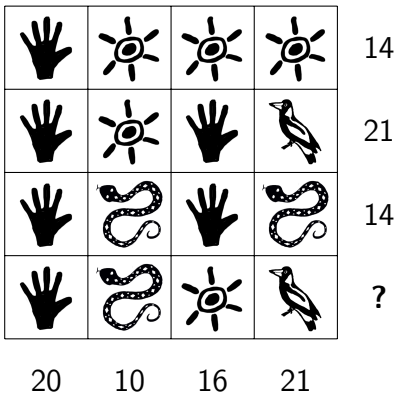
19. In this grid, the 4 symbols represent 4 different numbers. For most rows and columns, the total value is shown.
在右侧方格中，4 种符号分别代表 4 个不同的数字。如图所示，大多数行和列的总和已经标出。
What is the total value of the last row?
请问最后一行的总和是多少？



(A) 10 (B) 12 (C) 14 (D) 16 (E) 18

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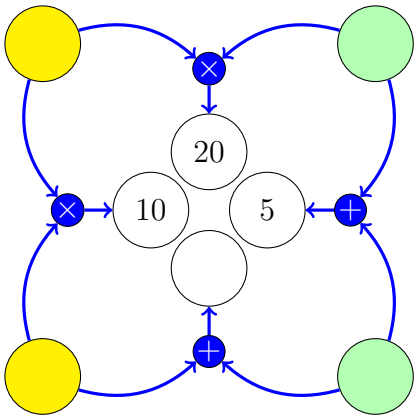
20. It takes 6 people 8 hours to harvest 5 truck-loads of mangoes. How long would it take 3 people to harvest 10 truck-loads of mangoes?
已知 6 个人 8 小时可以收获 5 车芒果，请问 3 个人收获 10 车芒果需要多长时间？

(A) 16 hours 16 小时 (B) 24 hours 24 小时 (C) 32 hours 32 小时
(D) 36 hours 36 小时 (E) 64 hours 64 小时

Questions 21 to 25, 5 marks each
21-25 题，每题 5 分

21. When this number puzzle is filled in, the 8 large circles will each contain a whole number greater than 0. The 4 inner numbers will be made from the 4 outer numbers using the arrows and the operations in the small blue circles.
请根据以下规则填写数字谜题：8 个大圆圈内均需填入大于 0 的整数；4 个内部数字则通过箭头和蓝色小圆圈内的运算符号，由 4 个外部数字运算得出。



For example, multiplying the numbers in the yellow circles on the left will give 10, and adding the numbers in the green circles on the right will give 5. How many ways are there to correctly fill in this number puzzle?
例如，左侧黄色圆圈内数字相乘结果为 10，右侧绿色圆圈内数字相加结果为 5。请问该数字谜题共有多少种正确的填写方式？

(A) 0 (B) 1 (C) 2 (D) 3 (E) 4

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22. Etienne has four square cards, each with a hole of a different shape:
小艾有四张正方形卡片，每张卡片上都有一个不同形状的孔：

Etienne picks two different cards, rotates them randomly and places one on top of the other. This combined shape is also a square with a hole. Which of the following **cannot** be that shape?
他挑选了两张不同的卡片，将其随机旋转后叠放在一起，组合后的图形仍然是一个带孔的正方形。请问以下哪个形状**不可能**是组合后的结果？

(A) (B) (C) (D) (E)

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23. Four local schools play in a football competition. Each school has one team that plays each of the other schools twice, once at their home ground and once at the other team's home ground. Then the two most successful teams play a grand final to decide the champion. How many matches are played in total?
四所地方学校参加足球竞赛。每所学校派出一支球队，与其他学校的球队进行两场比赛，一场在主场，一场在客场。之后，成绩最好的两支球队将进行一场总决赛以决出冠军。请问总共需要进行多少场比赛？

(A) 4 (B) 7 (C) 8 (D) 9 (E) 13

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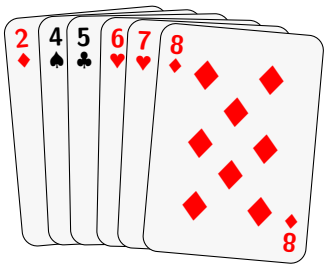
24. Carly has six cards with the numbers 2, 4, 5, 6, 7 and 8 on them. Using these cards, she can make 30 different two-digit numbers. For example, the smallest number she can make is 24 and the largest is 87. How many of these two-digit numbers are multiples of 3?
小莉有六张数字卡片，分别标有数字 2、4、5、6、7、8。
她用这些卡片可以组成 30 个不同的两位数。例如，最小的两位数是 24，最大的两位数是 87。请问在这些两位数中，有多少个是 3 的倍数？



(A) 7 (B) 10 (C) 11 (D) 12 (E) 15

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25. A tractor can pull three times as much weight as a horse can. A horse can pull half as much weight as a ute can.
一台拖拉机能拉动的重量是一匹马的三倍。
一匹马能拉动的重量是一辆小货车的一半。
Which of the following can pull the most combined weight?
请问以下哪个选项可以拉动最大的总重量？

(A) 1 horse and 4 utes　　1 匹马和 4 辆小货车
(B) 1 tractor and 1 ute　　1 台拖拉机和 1 辆小货车
(C) 2 horses and 2 utes　　2 匹马和 2 辆小货车
(D) 1 tractor and 4 horses　　1 台拖拉机和 4 匹马
(E) 1 tractor, 1 horse and 1 ute　　1 台拖拉机、1 匹马和 1 辆小货车

For questions 26 to 30, shade the answer as a whole number from 0 to 999 in the space provided on the answer sheet.
问题 26-30 的答案为 0-999 之间的整数，请将答案填涂在答题卡对应区域。

Questions 26–30 are worth 6, 7, 8, 9 and 10 marks, respectively.
第 26-30 题分别为 6, 7, 8, 9, 10 分。

26. Mrs Doublet has two pairs of twins. The first pair are both girls and she named them Alice and Bethan. Less than a decade later she had two boys: Yann and Zachary. This year, she was amazed to find that when she multiplied their four ages, the result was 2025. What is the sum of their ages?
杜太太育有两对双胞胎。第一对双胞胎是女孩，名叫丽丽和贝贝。十年内，她又生下了一对双胞胎男孩，名叫阳阳和章章。今年，杜太太惊讶地发现四个孩子的年龄相乘等于 2025。请问他们的年龄之和是多少？

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27. The reading on the dashboard of Karl's car shows that the car has travelled a total of 75 986 km.

He notices that the digits are five numbers in a row – in this case $5, 6, 7, 8, 9$ – but in a jumbled order. How many kilometres must Karl travel until the next time there are 5 numbers in a row, but jumbled up?

卡尔的汽车里程表显示已经行驶了 75 986 km。他注意到这些数字是连续的五个数字（$5, 6, 7, 8, 9$），但顺序是打乱的。请问卡尔需要再行驶多少公里，才能遇到下一组被打乱顺序的五个连续数字？

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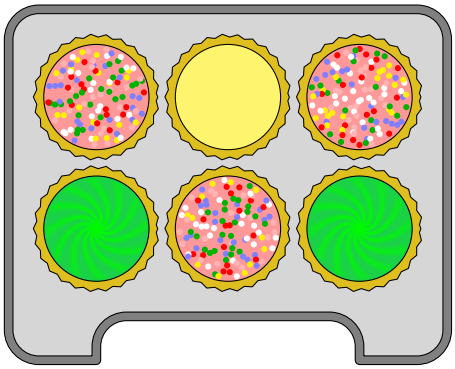
28. Dad has baked a tray of 6 cupcakes arranged in 2 rows of 3. Jenny's job is to do the icing for each cake. She has three different styles of icing: pink with hundreds and thousands, green swirl and plain yellow.

爸爸将烤好的 6 个纸杯蛋糕摆放成 2 行 3 列。珍珍需要为每个蛋糕装饰糖霜。她有三种不同款式的糖霜：粉色糖霜配彩色糖粒、绿色漩涡图案糖霜和纯黄色糖霜。



Dad wants them so that no two cakes next to each other in the same row or column have the same style of icing. The diagram shows one possibility. In how many ways can Jenny ice the cakes as Dad wants?

装饰规则为同一行或同一列中相邻的蛋糕不能使用相同款式的糖霜。图示展示了一种可能的装饰方案。在满足上述条件的情况下，请问珍珍有多少种装饰方案？

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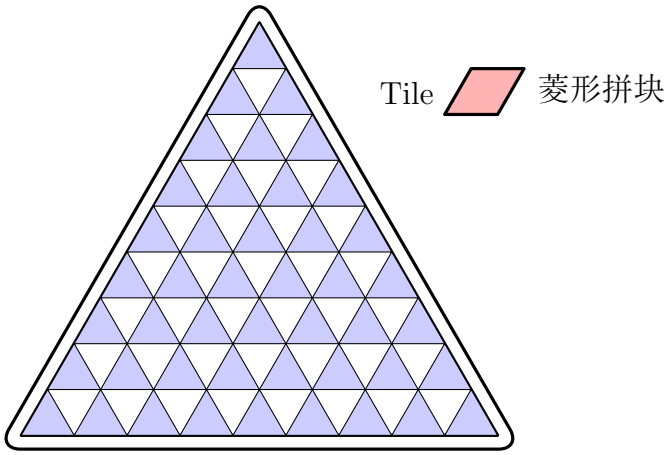
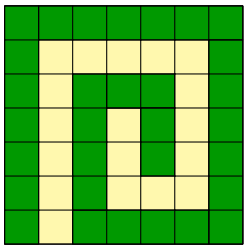
29. In the game of *Rhombus Blockade* a 1-by-1 rhombus tile must be placed on the triangular board shown so that it covers two grid triangles exactly.

The rhombus may be rotated. In how many different ways can the rhombus be placed?

在游戏 菱形迷阵 中，玩家需要将一个 1×1 的菱形拼块放置在图中的三角形游戏板上，使其恰好覆盖两个网格三角形。已知菱形拼块可以旋转，请问该菱形拼块有多少种不同的放置方法？



30. Jane is sewing a special patchwork quilt for 2025. The design has two spiral pathways one square wide that go from the centre to the edge. One is green and the other gold. The start of the design is shown, and it keeps spiraling out until there are 2025 squares in a 45-by-45 grid. In the finished 45-by-45 quilt, all edge squares except one will be green. What is the difference between the numbers of green and gold squares in the final quilt?



小简正在缝制一床特别的拼布被子，以纪念 2025 年。被子的设计包含两条螺旋路径，每条路径的宽度为一个方格，从中心向外螺旋延伸至边缘。其中一条路径为绿色的，另一条为金色。初始设计如图所示，螺旋持续向外扩展，直至形成一个 45×45 的网格，共计 2025 个方格。在缝制完成的被子上，除了一个边缘方格外，其余所有边缘方格都将是绿色。请问在最终缝制完成的拼布被子中，绿色方格与金色方格的数量相差多少？